

Thought for 25s >

Absolutely — before moving forward, this should be documented clearly. Below is a clean **ready-to-save Markdown documentation** for your Project 2.

Create this file:

</> PowerShell



```
notepad .\docs\REINVENT4_INSTALLATION_AND_SANITY_CHECK.md
```

Then paste this:

</> Markdown



```
# REINVENT4 Installation, Troubleshooting, and Sanity Check

Project: `FXA_REINVENT_Portfolio`
Environment: `fxa_reinvent4_py311`
Platform: Windows PowerShell
REINVENT version: `4.8.24`
PyTorch: `2.12.0+cpu`
Run mode tested: CPU sampling

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## 1. Project location

```powershell
cd C:\Users\eluma\Documents\mission_30days_3_portfolios\FXA_REINVENT_Portfolio
```

Final project structure used:

```
FXA_REINVENT_Portfolio/
├─ configs/
├─ data/
│ ├─ generated/
│ └─ reference/
├─ docs/
├─ external/
│ └─ REINVENT4/
├─ models/
├─ results/
│ ├─ metrics/
│ ├─ tables/
│ └─ figures/
└─ scripts/
```



## 2. Conda environment

The first attempt used Python 3.10, but the installed REINVENT4 package required Python 3.11 or newer. Therefore, a new environment was created.

PowerShell

```
conda create -n fxa_reinvent4_py311 python=3.11 -y
conda activate fxa_reinvent4_py311
python --version
where python
```



Expected environment:

```
fxa_reinvent4_py311
Python 3.11.x
```



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## 3. Git installation issue

Initial problem:

```
git : The term 'git' is not recognized
```



Fix:

PowerShell

```
conda activate fxa_reinvent4_py311
conda install -c conda-forge git -y
git --version
```



Reason: REINVENT installation later needed Git to install the `iSIM` dependency from GitHub.

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## 4. Clone REINVENT4

PowerShell

```
cd C:\Users\eluma\Documents\mission_30days_3_portfolios\FXA_REINVENT_Portfolio

New-Item -ItemType Directory -Force external
cd external
```



```
git clone --depth 1 https://github.com/MolecularAI/REINVENT4.git
cd REINVENT4
```

A Windows warning appeared about case-sensitive file collisions in tutorial map files. This did not block the core REINVENT installation.

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## 5. Check installer help

PowerShell

```
python install.py --help
```



The installer showed:

```
usage: install.py [-h] [-d PKGS [PKGS ...]] [-e] [--dry-run] NAME
```

```
NAME: cpu, cu124, rocm6.2.4, mac, etc.
```



For this project, CPU installation was selected first for stability.

---

## 6. Dry run

PowerShell

```
python install.py cpu --dry-run *> ../../scripts\02_reinvent_install_dryrun_py311.log
Get-Content ../../scripts\02_reinvent_install_dryrun_py311.log -Tail 80
```



Dry run command shown:

```
pip install .[all,chemprop2] --extra-index-url https://download.pytorch.org/whl/cpu
```



## 7. Install REINVENT4

PowerShell

```
python install.py cpu *> ../../scripts\02_reinvent_install_py311_retry2.log
Get-Content ../../scripts\02_reinvent_install_py311_retry2.log -Tail 120
```



Successful install evidence:

Successfully built reinvent molvs iSIM

Successfully installed ... reinvent-4.8.24 ... torch-2.12.0+cpu ... torchvision-0.2



The install log confirmed that `reinvent-4.8.24`, `torch-2.12.0+cpu`, `torchvision-0.27.0+cpu`, `chemprop`, `OpenEye-toolkits`, and other dependencies were installed successfully.

Pasted text

## 8. Troubleshooting during install

### Problem A: Python 3.10 was incompatible

Error:

ERROR: Package 'reinvent' requires a different Python: 3.10.20 not in '>=3.11'



Fix:

PowerShell

```
conda create -n fxa_reinvent4_py311 python=3.11 -y
conda activate fxa_reinvent4_py311
```



### Problem B: Missing torch and RDKit

Errors:

ModuleNotFoundError: No module named 'torch'  
ModuleNotFoundError: No module named 'rdkit'



Fix:

PowerShell

```
conda install -c conda-forge rdkit pandas numpy scipy scikit-learn tqdm pydantic toml -y
python -m pip install torch --index-url https://download.pytorch.org/whl/cpu
```



### Problem C: Git missing during iSIM installation

Error:

ERROR: Cannot find command 'git'



Fix:

<> PowerShell



```
conda activate fxa_reinvent4_py311
conda install -c conda-forge git -y
```

Then rerun:

<> PowerShell



```
cd C:\Users\eluma\Documents\mission_30days_3_portfolios\FXA_REINVENT_Portfolio\external\
python install.py cpu *> ..\..\scripts\02_reinvent_install_py311_retry2.log
```

## Problem D: Windows `resource` module error

Error:

ModuleNotFoundError: No module named 'resource'



Reason: Python's `resource` module is Unix/Linux-specific. REINVENT imported it in `hw_report.py`.

Fix: patch `hw_report.py` to make `resource` optional.

<> PowerShell



```
@'
from pathlib import Path
import site

matches = []

for sp in site.getsitepackages():
 p = Path(sp) / "reinvent" / "utils" / "hw_report.py"
 if p.exists():
 matches.append(p)

if not matches:
 raise SystemExit("Could not find reinvent/utils/hw_report.py")

path = matches[0]
text = path.read_text(encoding="utf-8")
```

```

backup = path.with_suffix(".py.bak_windows_resource_patch")
if not backup.exists():
 backup.write_text(text, encoding="utf-8")

old = "import resource"

new = """try:
 import resource
except ModuleNotFoundError:
 class _DummyResource:
 RUSAGE_SELF = 0
 RUSAGE_CHILDREN = 0

 @staticmethod
 def getrusage(*args, **kwargs):
 class _Usage:
 ru_maxrss = 0
 return _Usage()

 resource = _DummyResource()
"""

if "class _DummyResource" in text:
 print("Already patched:", path)
else:
 if old not in text:
 raise SystemExit("Could not find 'import resource' line to patch.")
 text = text.replace(old, new)
 path.write_text(text, encoding="utf-8")
 print("Patched:", path)
 print("Backup:", backup)
'@ | python

```

## Problem E: Pillow / PIL DLL error

Error:

```
ImportError: DLL load failed while importing _imaging
```



Reason: `torchvision` imported `PIL.Image`, but the Pillow DLL was broken.

Fix:

PowerShell



```

python -m pip uninstall -y pillow PIL
conda remove pillow -y
python -m pip install --no-cache-dir --force-reinstall pillow

```

Then test:

</> PowerShell



```
python -c "from PIL import Image; print('PIL Image OK')"
```

## Problem F: Missing RDKit and matplotlib after Pillow cleanup

Warning:

```
reinvent 4.8.24 requires matplotlib<4,>=3.7, which is not installed.
reinvent 4.8.24 requires rdkit>=2025.09.1, which is not installed.
```



Fix:

</> PowerShell



```
python -m pip install --no-cache-dir --only-binary=:all: "rdkit>=2025.09.1" "matplotlib>=
```

## Problem G: Missing charset-normalizer

Error from `pip check`:

```
requests 2.34.2 requires charset-normalizer, which is not installed.
```



Fix:

</> PowerShell



```
python -m pip install charset-normalizer
python -m pip check
```

Final result:

```
No broken requirements found.
```



The final dependency check and import tests passed: `pip check` returned no broken requirements, and PIL, RDKit, torchvision, and REINVENT all imported correctly.

Pasted text

## 9. Final environment verification

Run from project root:

<> PowerShell



```
cd C:\Users\eluma\Documents\mission_30days_3_portfolios\FXA_REINVENT_Portfolio
conda activate fxa_reinvent4_py311

python -m pip check

reinvent --help *> .\scripts\03_reinvent_help_after_dependency_fix.log
Get-Content .\scripts\03_reinvent_help_after_dependency_fix.log -Head 120

python -c "from PIL import Image; print('PIL Image OK')"
python -c "from rdkit import Chem; print('RDKit OK', Chem.MolToSmiles(Chem.MolFromSmiles('CC1=CC=CC=C1')))"
python -c "import torchvision; print('torchvision', torchvision.__version__)"
python -c "import reinvent; print('REINVENT import OK')"
```

Expected output:

```
No broken requirements found.
REINVENT 4.8.24 using PyTorch 2.12.0+cpu
PIL Image OK
RDKit OK CCO
torchvision 0.27.0+cpu
REINVENT import OK
```



## 10. Download REINVENT prior model

The cloned REINVENT4 repository did not include a `priors` folder or `reinvent.prior`. This caused the first sampling run to fail with:

```
RuntimeError: model file ...\external\REINVENT4\priors\reinvent.prior is not access
```



The logs confirmed that `external\REINVENT4\priors\reinvent.prior` did not exist and the `priors` folder was missing.  Pasted text

Download the official prior:

<> PowerShell



```
cd C:\Users\eluma\Documents\mission_30days_3_portfolios\FXA_REINVENT_Portfolio
conda activate fxa_reinvent4_py311

New-Item -ItemType Directory -Force .\external\REINVENT4\priors
```



```
curl.exe -L "https://zenodo.org/records/15641297/files/reinvent.prior?download=1" -o ".\e
```

Verify:

<> PowerShell



```
Test-Path .\external\REINVENT4\priors\reinvent.prior

Get-Item .\external\REINVENT4\priors\reinvent.prior | Select-Object FullName,Length

Get-FileHash .\external\REINVENT4\priors\reinvent.prior -Algorithm MD5
```

Expected:

```
Test-Path = True
Length = 23226277
MD5 = F268EB072F4FCA69CA9434768D3CD461
```



## 11. Minimal REINVENT sampling config

Create:

<> PowerShell



```
@'
run_type = "sampling"
device = "cpu"
json_out_config = "results/metrics/03_sanity_sampling_config.json"

[parameters]

model_file = "external/REINVENT4/priors/reinvent.prior"
sample_strategy = "multinomial"
temperature = 1.0

output_file = "data/generated/reinvent_sanity_100.csv"

num_smiles = 100
unique_molecules = true
randomize_smiles = true
'@ | Set-Content .\configs\03_sanity_sampling.toml
```

Check:

<> PowerShell



```
Get-Content .\configs\03_sanity_sampling.toml
```

Important: do not type this line directly into PowerShell:

```
model_file = "external/REINVENT4/priors/reinvent.prior"
```



That line belongs inside the TOML config file, not as a PowerShell command. Typing it directly causes:

```
model_file : The term 'model_file' is not recognized
```



This happened during troubleshooting and was only a PowerShell usage mistake, not a REINVENT issue.  Pasted text

---

## 12. Run sanity sampling

<> PowerShell



```
reinvent -l .\results\metrics\03_sanity_sampling.log .\configs\03_sanity_sampling.toml
```

Inspect log:

<> PowerShell



```
Get-Content .\results\metrics\03_sanity_sampling.log -Tail 100
```

Inspect generated molecules:

<> PowerShell



```
Get-ChildItem .\data\generated
```

```
Import-Csv .\data\generated\reinvent_sanity_100.csv | Select-Object -First 10
```

---

## 13. Sanity sampling result

Final successful result:

```
Started REINVENT 4.8.24
Python version 3.11.15
PyTorch version 2.12.0+cpu
RDKit version 2026.03.3
Platform Windows
```



```
Using CPU
Starting Sampling
Writing sampled SMILES to CSV file data/generated/reinvent_sanity_100.csv
Sampling 100 SMILES from model external/REINVENT4/priors/reinvent.prior
Removed 1 invalid SMILES
Finished REINVENT
```

Generated file:

```
data/generated/reinvent_sanity_100.csv
```



The first generated molecules were successfully read with:

PowerShell



```
Import-Csv .\data\generated\reinvent_sanity_100.csv | Select-Object -First 10
```

## 14. Final status

REINVENT4 installation status:

```
Conda environment: fxa_reinvent4_py311
Python: 3.11.15
REINVENT: 4.8.24
PyTorch: 2.12.0+cpu
torchvision: 0.27.0+cpu
RDKit: 2026.03.3
PIL/Pillow: working
pip check: No broken requirements found
REINVENT CLI: working
Prior model: downloaded and checksum verified
Sanity sampling: completed
Generated output: data/generated/reinvent_sanity_100.csv
```



Conclusion:

REINVENT4 is installed and functional on Windows CPU mode after small Windows-specific fixes. The environment can now generate molecules from the official `reinvent.prior`. The next project step is to score generated molecules using the Project 1 Factor Xa scaffold-aware Random Forest model.



This documentation is solid for your portfolio because it records not only the final